

Miquel López Escoriza

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EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Sep. 2025 – May 2027

MSc in Data Science; GPA: 6/6

Lausanne, Switzerland

- Top grades: Markov Chains and Algorithmic Applications, Mathematics of Data, Machine Learning.

Universitat Politècnica de Catalunya (BarcelonaTech) – CFIS

Sep. 2020 – May 2025

Double BSc in Data Science and Engineering & Engineering Physics; GPA: 8.4/10 and 8.6/10

Barcelona, Spain

- Selected for CFIS, UPC's highly selective double-degree engineering program; acceptance rate below 10%.
- Honours: Mathematical Optimization, Probability, Discrete Mathematics, Calculus, Linear Algebra, Advanced Physics.

RELEVANT PROJECTS

Large Scale Multimodal Dataset Construction, [VILAB](#), [EPFL](#)

Feb. 2026 – Present

Multimodal learning, dataset curation, scalable data processing

Lausanne, Switzerland

- Working on the construction of a large-scale multimodal image and video dataset.
- Benchmarking state-of-the-art pseudolabelers for optical flow, depth, surface normals, SSL features and others.
- Developing a distributed GPU-accelerated pipeline to efficiently pseudolabel and tokenize large-scale image and video datasets.

Segmentation Foundation Model for 3D CT Segmentation, [BIG](#), [EPFL](#)

Sep. 2025 – Jan. 2026

Medical imaging, 3D segmentation, foundation models

Lausanne, Switzerland

- Adapted SAM-2 to 3D CT volumes using inference-time memory and propagation strategies.
- Evaluated on 2,500 CT scans from TotalSegmentator, achieving up to +12.5% Dice improvement on bone structures.

Foundation Models for Brain MRI, [NYU Courant Institute](#)

Sep. 2024 – May 2025

Medical imaging, self-supervised learning, 3D vision

New York, United States

- Bachelor thesis as a one year research internship under [Prof. Carlos F. Granda](#)'s supervision.
- Built a large-scale brain MRI dataset(100k) with aligned anatomical and clinical labels.
- Pretrained ViT models using MAE and DINO on this dataset.
- Designed a supervised segmentation pretraining stage to incorporate anatomical priors into image embeddings.
- Paper currently under review at [npj Digital Medicine](#).

ACHIEVEMENTS

Bachelor's thesis distinction: Awarded 10/10 with highest honors; recognized as the best bachelor's thesis in AI by the Faculty of Informatics and second best university-wide.

Research Scholarship: Awarded a triple scholarship by CFIS, NYU, and the Spanish Government to conduct my senior thesis at NYU Courant Institute.

Best national grade in university entrance exam: Achieved the best grade, 13.75/14, in the Andorran 2020 University Access Exam.

University scholarship: Awarded a scholarship covering tuition and housing costs for admission to CFIS.

TECHNICAL SKILLS

Programming Languages: Python, C++, R, SQL, Matlab

Relevant tools and libraries: PyTorch, TensorFlow, OpenCV, NumPy

LANGUAGE SKILLS

Spanish, Catalan: Native Speaker

English: Bilingual(C2 level)

French: B2 level

INTERESTS

Former competitive skier with 12 years of experience; also interested in surfing, backpacking, and outdoor travel.